

URBAN ENVIRONMENTAL PROBLEMS

This lecture notes enlightens you on Urbanization, help identify major problems that have surfaced due to the unprecedented rate of urban growth in India.

Urbanization refers to the population shift from rural to urban areas, the corresponding decrease in the proportion of people living in rural areas, and the ways in which societies adapt to this change.

Reasons for urbanization:

• Employment opportunities	Increase in population
• Better education for children	• Rural to urban migration
• Better commodity availability	• Better standards of living
• Better health care facilities	• Rural push factors and urban pull factors

Impacts of urbanization on the Environment:

1. **Habitat loss and deforestation:** Forest fires because of natural or manmade causes are devastating to the flora and fauna, resulting in the loss of biodiversity.

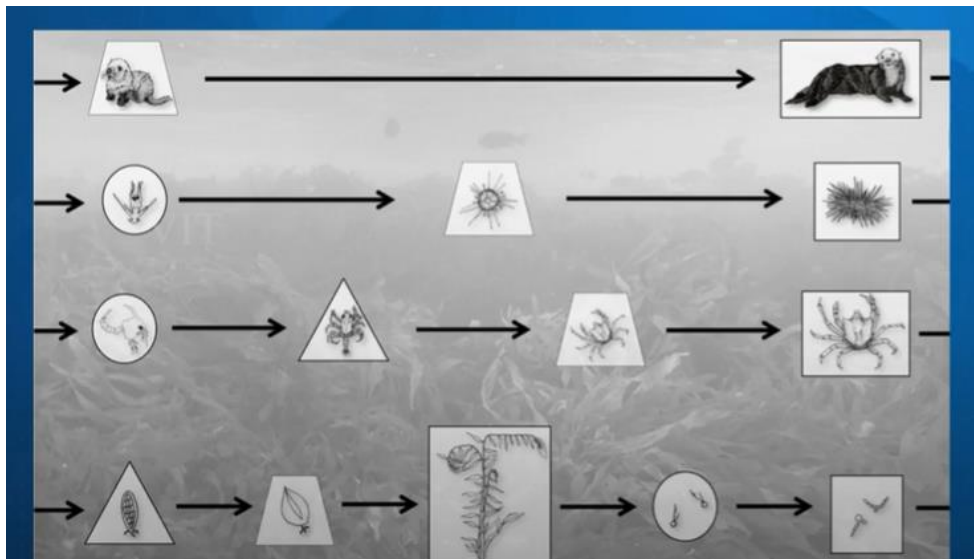


©CanStockPhoto.com-csp79280678

2. The evolution of lifecycle and traits

Urbanization alters abiotic and biotic environments over time and space. Several ecological changes are associated with urbanization and have the potential to strongly affect the evolution of the urban population

Therefore, urbanization can also have an impact on the evolution of life cycles and traits that help different species to survive and reproduce in disturbed or altered ecosystems. For example, some bird populations living in urban environments have altered their beak shape to be able to more effectively eat the seeds in human-made bird feeders.



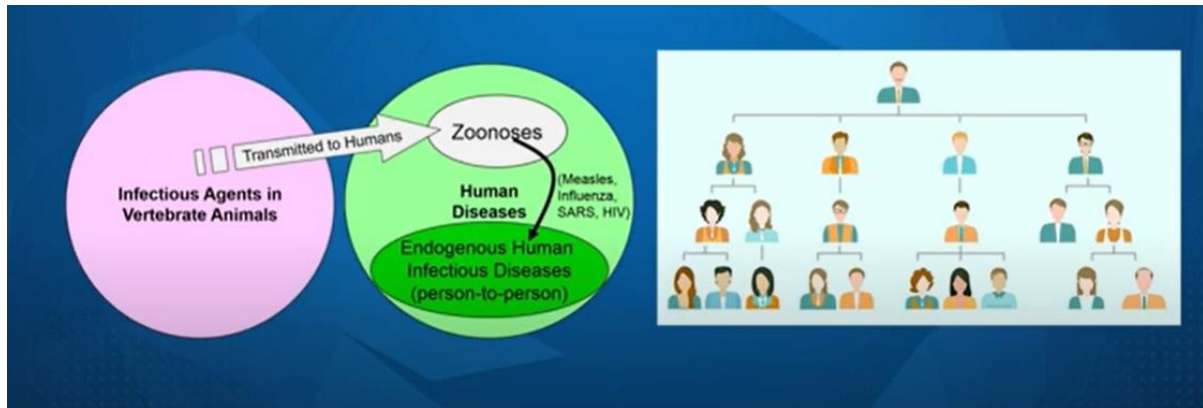
3. Transmission of diseases

The most predominant health problems associated with urbanization are poor nutrition, pollution-related health conditions, the emergence of infectious diseases, inadequate sanitation, and housing conditions, resulting in adverse health outcomes. They influence the individual's quality of life while also compromising on public health, placing burdens on public health systems and resources.

For example, the impact of urbanization on the transmission of disease can be clearly seen in the recent COVID-19 pandemic. In India, metropolitans contributed three-fourths of total cases from the beginning. The transport networks attributed significantly to transmitting the virus from the urban containment zones.

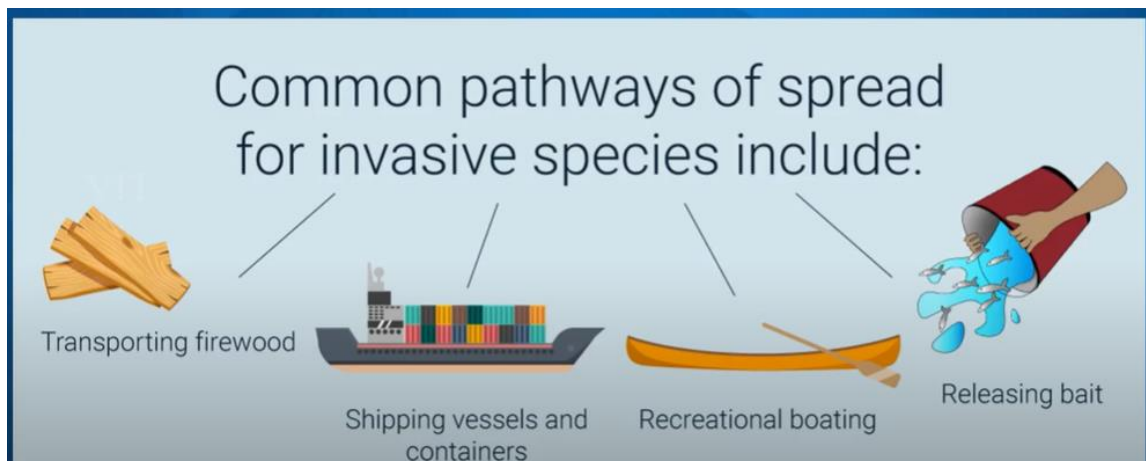
The outbreak of the COVID-19 epidemic has triggered the discussion of the relationship between urbanization and the spread of infectious diseases. One hypothesis attributes the intensified transmission of COVID-19 to the “urban diseases” caused by the rapid urban expansion, such as increased population ratio, dense buildings, environmental pollution, and deteriorated sanitation by many people, which finally threatens residents’ public health. The

main basis of this hypothesis is that there is a significant gap in the number of confirmed COVID-19 cases between urban and rural areas in various cities.



The above figure shows that because of urbanization, the people live much more closure in the small apartments/houses and a little sneeze or a cough can easily and very quickly transmit the disease (COVID-19) to a large population in a short- duration of time.

4. Increasing the transport of invasive species



What Makes a Species "Invasive"?

An invasive species can be any kind of living organism—an amphibian (like the cane toad), plant, insect, fish, fungus, bacteria, or even an organism's seeds or eggs—that is not native to an ecosystem and causes harm. They can harm the environment, the economy, or even human health. Species that grow and reproduce quickly, and spread aggressively, with the potential to cause harm, are given the label “invasive.”

How Invasive Species Spread

Invasive species are primarily spread by human activities, often unintentionally. People, and the goods we use, travel around the world very quickly, and they often carry uninvited

species with them. Ships can carry aquatic organisms in their ballast water, while smaller boats may carry them on their propellers. Insects can get into the wood, shipping palettes, and crates that are shipped around the world. Some ornamental plants can escape into the wild and become invasive, while some invasive species are intentionally or accidentally released to pets.

Threats to Native Wildlife

Invasive species cause harm to wildlife in many ways. When a new and aggressive species is introduced into an ecosystem, it may not have any natural predators or controls. It can breed and spread quickly, taking over an area. Native wildlife may not have evolved defenses against the invader, or they may not be able to compete with a species that has no predators.

5. Paving the land with concrete



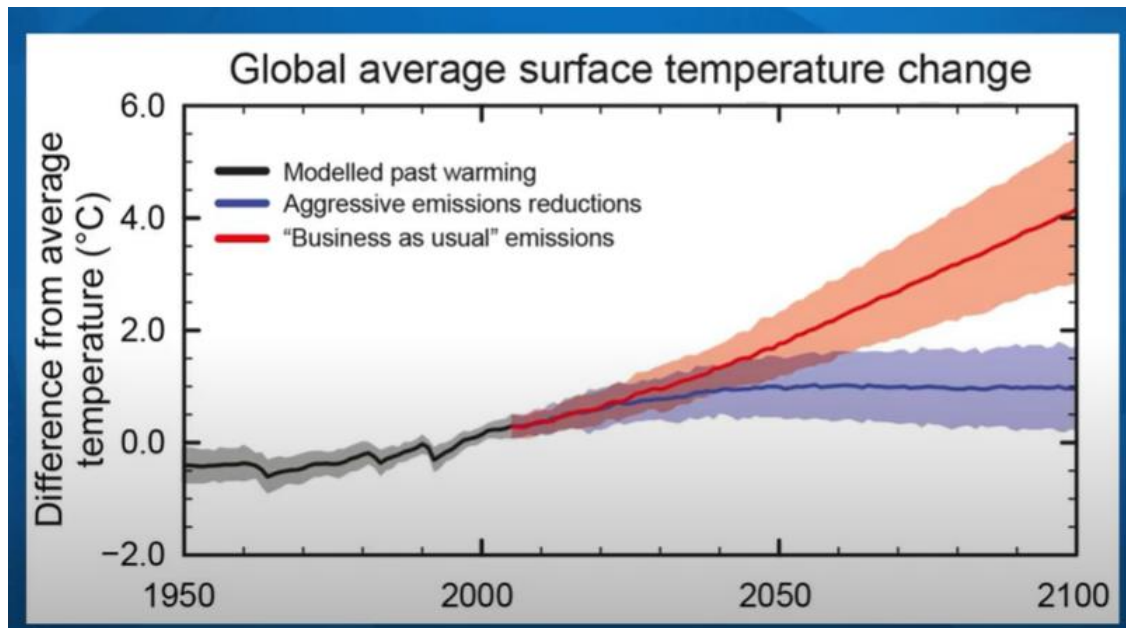
Paving land with concrete has a huge impact on the ecosystem such as:

- Increase in water runoff
- Increase in erosion
- Decrease in soil quality
- Decreases water quality by increasing sediment and pollutants in rivers and streams.

6. Increase in the regional temperature

As cities add roads, buildings, industry, and people, heat islands are created in urban areas. An urban heat island is a name given to describe the characteristic warmth of both the

atmosphere and surfaces in urban sprawls as compared to their non-urbanized or less urbanized surroundings



In the above figure with the time frame 1950-2100, there is Modelled past warming followed by Aggressive emissions reductions and then Business as usual emissions.

Impact of high temperature on human health and environment:

- Drinking lots of water to lower the body temperature and prevent dehydration.
- Vitamin D deficiency because of non-exposure to sunlight and avoiding high temperature
- Occurrence of several new diseases
- Plant growth, yield, and harvesting time have been changed and affected

7. Other impacts of urbanization:

- Industrial growth
- Increase in employment
- Social factors (moving more people from rural to urban)
- Economic problem during epidemic/pandemic
- Political turmoil
- Modernization
- Increase in population density
- Closely constructed houses
- Increased chance of epidemics

- Increased traffic resulting in traffic jams
- More need of resources- Energy, water, and fuel, stress has been increased on locally available resources
- High level of pollution in the air, water, soil-waste dumping
- Decreased aesthetic appeal of the landscape
- Loss of farmland
- Reduced species diversity
- Increased storm water runoff, because of the construction of pavement. As paving land with concrete can increase water runoff, increase erosion and decrease soil quality.
- Increased risk of flooding
- Excessive removal of native vegetation
- Ecosystem of fragmentation

8. Positive impacts of Urbanization

- More employment opportunities
- Access to recreation, ecotourism,
- Consumerism

Sustainable Urban Development: is important because urban areas contribute significantly to the Gross Domestic Product (GDP). They contribute increasingly to export and is a rich place for capital formation. Cities offer quality education and health care; arts and science; technology and innovation and transport and communication.

- Some of the pre-requisites of sustainable urban development are:
- Income and output to be produced at a constant or even increasing return to scale.
- City should assure a minimum level of living to its inhabitants.
- A shift in the attitude of the people in the direction towards enterprise and equity.
- City should become self-reliant and sustain itself without much depending on external sources.
- The developmental institutions undergo a continual renewal to maintain their relevance to the needs of the urban areas.
- Participation of the private sector, NGOs, and CVOs should be encouraged in order to maintain the level of development.

- Management of Urban growth to promote minimal use of environmental capital, while meeting social and economic goals.
- Zero waste disposal to be adopted and to become a habit. Japan leads the world in zero waste management. Indore, an Indian city has bagged the first prize for five consecutive years for their waste management.

The Agenda 21 of WCED held in 1992 in Rio-de-Janerio for promoting Sustainable Human Settlement Development is

- Providing adequate shelter to all
- Improving human settlements management
- Promoting Sustainable land use planning and management.
- Promoting the integrated provision of environmental infrastructure: water, sanitation, drainage, hazardous and solid waste management.
- Promoting sustainable energy and transport systems in human settlements.
- Promoting human settlement planning and management in disaster-prone areas.
- Promoting sustainable construction industry activities.
- Promoting human resources development and capacity building for human settlement development

9. Pros and Cons of Urbanization

Pros of Urbanization	Cons of Urbanization
Better public transport	Higher level of air pollution
Better overall infrastructure	More particles pollution
More education opportunities	Noise pollution
Increasing property prices of homeowners	Light pollution
People can use their time more efficiently	Littering
More conveniently if you have kids	Cities may become quite crowded

Connection to the electricity grid and sewer system	Traffic jams
Better internet connection	Higher level of stress
More efficient in terms of land use	Increasing health issues
People in cities are usually more tolerant	Increasing rent for tenants
Better connection of healthcare facilities	Higher chances of homelessness
Better R&D opportunities to foster technological progress	Development of slums
Urbanization attracts tourists	Spread of diseases
Multicultural attitude	Waste management of problems
Easier to socialize	Pets may not like the city life
Lower Unemployment rates	Lack of natural spaces

10. Summary

We understood urbanization, the reasons for urbanization and its problems, the impact of urbanization on the environment, discuss the problems of housing, water supply, transport, and environmental pollution in urban India, and the association of urbanization with human development and progress. Let's ensure to consciously reduce the problems of urbanization using advanced technical and apt solution.